Candidate Seat No _______________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Bachelor of Computer Applications
Semester-IV University Examination May, 2018
CA205/CA221 – Data Structures and Algorithms
Date: 08-05-2018, Tuesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 70

Instructions:

1. **Attempt both sections in separate answer books.**
2. **Figure to right indicates marks.**
3. **Make suitable assumption when necessary.**

SECTION-I

Q – 1 Do as directed. **[11]**

1 What do you mean by composite Data Structure?
2 What is the functionality of malloc()?
3 What is the main advantage of dynamic memory allocation?
4 Define Term : Data Structure
5 What do you mean by circular and non-circular Data Structure?
6 What do you mean by abstract Data Structure?
7 What do you mean by row major and column major in 2 - D array?
8 int k = 10,*p; p = &k; printf("%__",*p); Fill in the blank and print the output also.
9 What is structure? Explain its significance in context of array.
10 What is the difference between "a" and 'a'?
11 What will be the size of a node which contains one float data and pointer to next member?
A. 4 Bytes B. 6 Bytes
C. 8 Bytes D. None of these

Q – 2 (A) Write a C function to insert after and before in double link list. **[8]**

Q – 2 (B) Explain difference between queue and circular queue. **[4]**

OR

Q – 2 (A) Write a C function to delete first node of single link list and delete a specific node in the single link list. **[8]**

Q – 2 (B) (A+B)*(C^D)/T – Convert it into prefix notation. **[4]**

Q – 3 Attempt the following. (Any 2) **[12]**

1. Consider one integer array with 410 rows and 120 columns. Address of [0,0] is 1250. Find out address of [230,80] with row major and column major technique. Take [0,0] as base.
2. Write down algorithm for Push and Pop operation in stack.
3. Differentiate: Static Memory Allocation & Dynamic Memory Allocation
Differentiate: Link List & Stack
4. Explain time and space complexity of algorithm with suitable example.

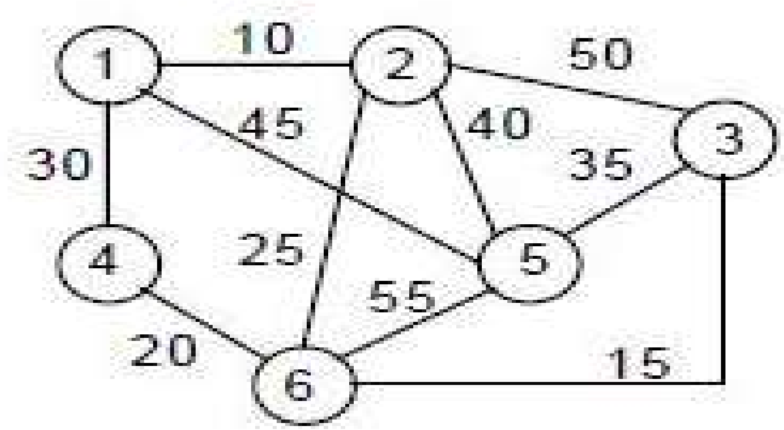
SECTION-II

Q – 4 Do as directed. [11]

1. What is the formula for parent, left child and right child in max heap?
2. List out prerequisite condition for merge sort and Binary Search.
3. What do you mean Divide & Conquer technique?
4. What do you mean by blind search?
5. What do you mean by degree in tree?
6. Define: Graph
7. Define: Leaf Node
8. Define: Level of the tree
9. Define: Connected Graph
10. Define: Directed Edge & Undirected Edge
11. A binary tree is always balanced binary tree. (True/False)

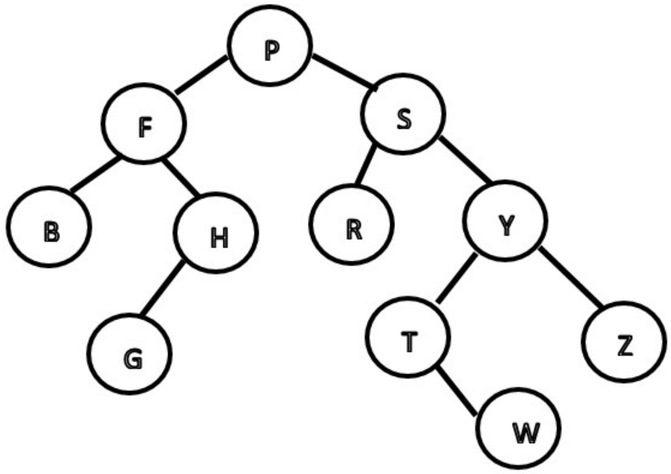
Q – 5 Attempt the following in detail. (Any 2) [12]

1. Convert the following into max heap and min heap.
[20,32,21,14,24,8,54,23,15,19,44,25]
2. Consider following array and apply steps of binary search to search value 8.
[20,32,21,14,24,8,54,23,15,19,44,25]
3. What is merging? Explain merging logic with suitable example. (Take 2 array with data and fill it into 3rd array)
4. Find out shortest path using Dijstra's Algorithm. Consider shortest path between node 1 – 3.

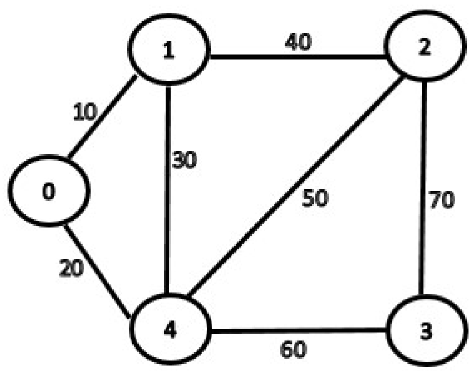


Q – 6 Attempt the following in detail. (Any 2) [12]

1. Traverse the following tree with Pre-order, In-order and Post-order.



2. Find out minimum spanning tree using Kurshkal's Algorithm.



3. Construct binary tree from given traversing.
   In-order: B D C K E A H J G I F
   Pre-order:A B C D E K F G H J I
4. Apply Quick Short on following given data.
   [20,32,21,14,24,8,54,23,15,19,44,25]
